

## **Statistical Programming Lab- ADL47**

### **Part A**

4. Write a R program to extract first 10 English letters in lower case and last 10 letters in upper case and extract letters between 22nd to 24th letters in upper case.
5. Write a R program to create an array of two 3x3 matrices each with 3 rows and 3 columns from two given vectors. Print the second row of the second matrix of the array and the element in the 3rd row and 3rd column of the 1st matrix.

### **Part B**

#### **Ex: 3: Use of Arrays and Functions**

You are developing an inventory management system for a small store. The system needs to handle inventory items and their quantities. Write a program that uses arrays to store inventory items and their quantities, and includes functions to add new items, update quantities, and display the inventory.

- Define an array to store inventory items.
- Define an array to store corresponding quantities.
- Implement functions to:
  1. Add a new item along with its quantity.
  2. Update the quantity of an existing item.
  3. Display the inventory items and quantities.

O Use the functions to manage the inventory and handle user interactions.

#### **Ex: 4 – Dataframes and dplyr**

- You are working as an educational analyst and need to analyze the performance of students in a school. You have data on student names, their scores in different subjects, and attendance. Write a program that uses data frames to manage and analyze student data, including calculating average scores, identifying students with low attendance, and generating a report.
- Create a data frame to store student information with columns: "Name", "Math\_Score", "Science\_Score", "History\_Score", "Attendance".
- Implement functions to:
  1. Calculate the average scores for each student.
  2. Identify students with attendance below a certain threshold.
  3. Generate a report with student names, average scores, and attendance status.

Use the functions to analyse student performance and generate the report